

This project is about Maven Fuzzy Factory. Main Business Scope: E-Commerce Sales

The company tracks:

- Total Revenue: 1.85M
- Net Profit: 1.16M
- Total Orders: 32K
- Total Sessions: 473K
- CVR % (Conversion Rate): 6.83%
- Refund % Rev: 4.61%
- AOV (Average Order Value): 57.35

The goal of this project:

- Monitor revenue & profit growth by year
- Track conversion rate and refund trends
- Analyze traffic sessions vs orders
- Compare product performance (e.g. Mr. Fuzzy, Love Bear, Sugar Panda)
- Evaluate marketing campaigns impact (nonbrand, brand, desktop targeted, pilot)

👉 The dashboard helps managers understand performance drivers and make better decisions.

2. Dataset

The main table is the Orders (Fact Table),

where each row represents one customer order.

➤ Tables: -

- Orders Table: Records order details (Order ID, Created at, Website session ID, User ID, Primary Product ID, Items Purchased, Price USD, Cogs USD)
- Order Items Table: Breaks down products within each order (Order item ID, created at, Order ID, Product ID, Its Primary Items Price USD, Cogs USD)
- Order Item Refunds Table: Tracks refunds (Order Item Refund ID, Created at, Order Item ID, Order ID, Refund amount USD)
- Products Table (Product ID, Created at, Product Name)
- Website Sessions Table (Website Session ID, created at, User ID, Is Repeat Session, Utm Source Utm Campaign, Utm Content, Device Type, Http Referer)
- Website Pageviews Table (Website pageview ID, Created at, Website Session ID, Page View URL)
- Calendar Table: when the sale happened (Year, Quarter, Month, Day)

👉 This dataset powers dashboards that help managers understand financial and marketing performance and make better decisions

3. Data Preparation:

Data Cleaning:

Handle all Missing values (null)

Remove duplicates

Splitting Columns

Correct data type issues

Rename columns

Make conditional Columns (Customized)

👉 This makes the data clean and stable

2. Enhancements

In this step, we improve the data for better analysis

👉 We add new columns and logic

Added:

- Day column → to analyze data by day
- Time column → to analyze hours (morning, evening, etc.)
- Target column → based on a parameter (can be changed easily)

👉 This helps us:

- Analyze Seasonality better
 - Track performance vs target
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4. Data Modeling

We used a Star Schema model

Tables:

- Fact Table → Orders
- Dimension Tables:
 - Order items
 - products
 - order item refunds
 - Website Page view
 - Website Sessions
 - Calendar

Relationships :

- Orders → Order (Order ID)
- Orders → Website sessions (Website sessions ID)
- Orders Items → product (product ID)
- Orders Items → Order items refunds (Order items id)
- Website sessions → website pageviews (Website sessions ID)
- Calendar → All tables

The Calendar table is a date dimension, linked to Fact table's Date field for time-based analysis.

👉 This helps:

- Fast performance
 - Easy analysis
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5. Calculation (DAX)

We divide the analysis into 3 main parts:

1. Revenue KPIs

- Total Revenue 👉 Goal: track total sales revenue

Measures used:

Total Revenue = CALCULATE (SUM (order_items[Price \$]),order_items[Refund state] ="sold")

- Total Order 👉 Shows how many orders we have

Measures used:

Total Orders = DISTINCTCOUNT (orders [Order_ID])

- Total Cost 👉 Shows total spending/costs

Measures used:

Total Cost = CALCULATE (SUM (order_items[Cost \$]),order_items[Refund state]="sold")

- Total Items Sold 👉 Goal: track total quantity sold

Measures used:

Total Items Sold = COUNTROWS (order_items)

- Total Refund Amount 👉 Goal: track refunded money

Measures used:

Total Refund Amount = SUM (order_item_refunds[Refund_amount \$])

- Sold Count 👉 Goal: track number of sold products

Measures used:

Sold Count = CALCULATE (COUNTROWS (order_items),order_items[Refund state] = "sold")

- Refunded Count 👉 Goal: track number of refunded orders

Measures used:

Refunded Count = CALCULATE(COUNTROWS(order_items),order_items[Refund state] = "refunded")

- Net Profit 👉 Goal: track actual profit after costs

Measures used:

Net Profit = [Total Revenue] - [Total Cost]

- Average Order Value 👉 Shows the average amount customers spend in one order

Measures used:

Average Order Value = DIVIDE ([Total Revenue], [Total Orders])

- Avg Items per Order 👉 Shows the average number of items in each order

Measures used:

Avg Items per Order = DIVIDE ([Total Items Sold], [Total Orders])

- AVG Price 👉 Shows the average price of sold items

Measures used:

AVG Price = AVERAGE (order_items[Price \$])

2. Website Performance KPIs

- **Frequent Visitor** 🖱️ Shows number of users who visited multiple times

Measures used

**Frequent Visitor = CALCULATE (COUNTROWS
(website_sessions),website_sessions[Repeated Session] = "FrequentVisitor")**

- **New Users:** 🖱️ Shows number of first-time visitors

Measures used

New Users = CALCULATE (COUNTROWS(website_sessions),website_sessions[Repeated Session] "NewComer")

- **Total Pageviews** 🖱️ Goal: track total viewed pages

Measures used

Total Pageviews (COUNTROWS (Website_ pageview)

- **Total Sessions** 🖱️ Goal: track total website visits

Measures used

**Total Sessions = CALCULATE (COUNT (website_sessions[website_session_id]),
USERRELATIONSHIP('Calender'[Date],'website_sessions'[Date]))**

- **Pages per Session:** 🖱️ Shows average number of pages viewed in one session

Measures used

**Pages per Session = CALCULATE (DIVIDE ([Total Pageviews], [Total Sessions], 0),
ALL (website_pageviews[Pageview_URL]))**

- **Nonconverted Sessions:** 🖱️ Shows the number of sessions without conversion/orders

Measures used

$$\text{Non Converted Sessions} = [\text{Total Sessions}] - [\text{CVR session}]$$

- **Repeated Views:** 🖱️ Shows repeated/extra pageviews made by users during sessions

Measures used

$$\text{Repeated Views} = [\text{Total Pageviews}] - [\text{Total Sessions}]$$

3. Rates %

- **CVR %:** 🖱️ Shows the percentage of sessions that resulted in orders/conversions

Measures used:

$$\text{CVR \%} = \text{VAR TotalOrders} = [\text{Total Orders}]$$

$$\text{VAR ForcedSessions} = \text{CALCULATE}(\text{COUNTROWS}('website_sessions'),$$

$$\text{USERRELATIONSHIP}('Calender'[Date], 'website_sessions'[Date]))$$

)

RETURN

$$\text{DIVIDE}(\text{TotalOrders}, \text{ForcedSessions}, 0)$$

- **CVR Session** 🖱️ Shows the percentage of sessions that converted into orders/actions

Measures used:

$$\text{CVR session} = \text{CALCULATE}([\text{Total Sessions}], \text{website_sessions}[\text{Session_Converted}] = "Y")$$

- **Profit Margin %:** 🖱️ Shows the percentage of profit earned from total sales revenue

Measures used:

Profit Margin % = $\text{DIVIDE}([\text{Net Profit}], [\text{Total Revenue}], 0)$

- **Refund % of Revenue:** 👉 Shows the percentage of revenue lost through refunds

Measures used:

Refund % of Revenue = $\text{DIVIDE}([\text{Total Refund Amount}], [\text{Total Revenue}])$

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- **Refund Rate %:** 👉 Shows how many orders had refunds compared to total orders

Measures used:

Refund Rate % =

$\text{DIVIDE}(\text{DISTINCTCOUNT}(\text{order_item_refunds}[\text{Order_ID}]), \text{DISTINCTCOUNT}(\text{orders}[\text{Order_ID}]), 0)$

- **Refund Status:** 👉 Shows whether the refund rate is acceptable or exceeds the allowed limit

Measures used:

Refund Status = $\text{IF}([\text{Refund Rate \%}] > 0.03, \text{"Above Limit"}, \text{"Within Limit"})$

- **Refund Target:** 👉 Sets the maximum allowed refund rate

Measures used:

Refund Target = 0.03

4. Time Intelligence

- Previous Month Revenue 🖱 Shows revenue from the previous month

Measures used:

Previous Month Revenue = CALCULATE ([Total Revenue], DATEADD(Calendar[Date], -1, MONTH))

- MoM %: 🖱 Shows monthly growth or decline percentage

Measures used:

MoM % = DIVIDE ([Total Revenue] - [Previous Month Revenue], [Previous Month Revenue])

- Previous Year Revenue 🖱 Shows revenue from the previous year

Measures used:

Previous Year Revenue = CALCULATE ([Total Revenue], SAMEPERIODLASTYEAR(Calendar[Date]))

- YoY % 🖱 Shows yearly growth or decline percentage

Measures used:

YoY % = DIVIDE([Total Revenue] - [Previous Year Revenue], [Previous Year Revenue])

6. Dashboard Layout

👉 Goal: make dashboard clear and easy to use

The dashboard is divided into 3 main pages:
(Orders – Sales – Performance)

👉 Each page focuses on one business area

General Flow

User reads each page like this:

1. KPI (What happened?)
2. Trend (How it changed over the Time?)
3. Drivers/Break Downs (Why it happened?)

Each Page analyze the KPI Main Drivers as following:

1 Overview Page

👉 Goal: a quick overview of overall performance

It includes:

- Total Sales
 - Total Orders
 - Conversion Rate
 - Key KPI Cards
 - Quick performance trend
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2 Financial Page

 **Goal: financial analysis (revenue – profit – losses)**

It includes:

- **Total Revenue**
 - **Net Profit**
 - **Profit Margin**
 - **Refund Impact on Profit**
 - **Month / Year comparisons**
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3 Orders Page

 **Goal: order analysis**

It includes:

- **Total Orders**
 - **Order Status (Completed / Cancelled / Returned)**
 - **Average Order Value**
 - **Top Products**
 - **Repeat Orders**
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4 Marketing & Traffic Analysis

 **Goal: understanding visitors and traffic sources**

It includes:

- **Total Sessions**
- **Traffic Sources (Ads / Organic / Social)**
- **Conversion by Channel**
- **Non-converted Sessions**
- **Funnel Analysis**

5 Refunds Page

👉 Goal: monitoring refunds

It includes:

- Refund Rate %
 - Refund Amount
 - Refund Status (Above / Within limit)
 - Refund reasons (if available)
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6 Performance / Decomposition Tree

👉 Goal: deep analysis of underlying drivers

It includes:

- Performance breakdown by:
 - Product
 - Region
 - Channel
- Root cause analysis
- Why performance is high/low

👉 This structure helps users understand:

- Where the problem is
 - What is causing the change
 - Who is responsible for the result 🚀
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7. Interaction

👉 Goal: give the user flexibility to explore data

Added features:

- Slicers → allow the user to filter data easily (ProductName, Year, Device type)

👉 This helps the user focus on specific data

- Parameter (Total revenue / Net Profit) → allow switching between different financial views
- Parameter (Refund % of Revenue / Refund Rate %) → It allows the user to switch between two different ways of displaying refund data in the same visual.

👉 This gives dynamic analysis in one visual

👉 These interactions make the dashboard dynamic and user-friendly 🚀

8. Conclusion

👉 Goal: support daily decisions and business Monitor

This dashboard helps managers to:

- Monitor daily sales and orders
 - 👉 Know if business is running well or not
- Track profit and performance
 - 👉 Understand if the company is making real profit

- Compare actual vs target
 - 👉 See if we are achieving goals
 - Identify problems quickly
 - 👉 Find low performance (product, campaign, source)
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👉 In simple words:

This dashboard turns data into clear actions

👉 Result:

Better decisions, faster response, and better business growth 🚀